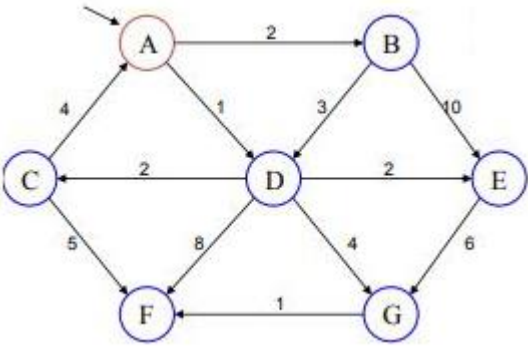


M.Sc. in CSE and IICT Admission Test in BUET - 2026**Model Test****Written****Total Marks – 100**

SI	Details	Marks
1	a) Let a hacker want to crack a password of a user from ABC website. The password length can be of 8 to 12 character long, containing alpha numeric characters. The password expires in every 30 days so that user has to renew it. Now mention the number of password the hacker may try in every second to crack the password. b) Find reverse polish notation from given expression using stack - $6\ 5\ 3\ * + 15\ 5\ / 7 +$ c) Discuss different conditions for occurring deadlock in OS	4 3 3
2	a) Using Dijkstra's Algorithm find the distance from node A to every node from the given graph below  b) Discuss how can we detect a negative cycle or negative edge is present within a given graph. c) A bag contains 4 red balls, 6 blue balls and 8 pink balls. One ball is drawn at random and replaced with 3 pink balls. A probability that the first ball drawn was either red or blue in colour and the second drawn was pink in colour?	4 3 3
3	a) Discuss the worst case time complexity of Quick sort mentioning proper recurrence relation b) Draw a max heap from the given data – 5, 6, 18, 4, 9, 13, 21, 16, 3, 14 c) Discuss the difference between run time and compile time polymorphism	5 3 2
4	a) Students(rollno: integer, sname: string) Courses(courseno: integer, cname: string) Registration(rollno: integer, courseno: integer, percent: real) Find the distinct names of all students who score more than 90% in the course numbered 107. b) Suppliers(sid:integer, sname:string, city:string, street:string) Parts(pid:integer, pname:string, color:string) Catalog(sid:integer, pid:integer, cost:real) Find the names of all suppliers who have not supplied only blue parts. c) Movie(mov_id, mov_name, rating, desc) Write a query to select only the movie names where the ratings are greater than 7 but less than 9. d) What are the possible errors can happens when we want to delete a column from a database table that has foreign reference in another table? Discuss how can we solve this problem.	2.5 2.5 2 3
5	a) Fill the gap with proper codes <code>int main()</code>	

	<pre> { int num = 150, sum = 0; for({ --- } Cout<< sum; // 6 } </pre> b) In the next World cup of cricket there will be 12 teams, divided equally in two groups. Teams of each group will play a match against each other. From each group 3 top teams will qualify for the next round. In this round each team will play against each others once. Four top teams of this round will qualify for the semifinal round, where they play the best of three matches. The Minimum number of matches in the next World cup will be c) Write a computer program that can be used to show following binary tree and give the sum of all the nodes. <pre> 5 / \ 4 7 / \ / \ 9 11 13 20 </pre> 24	
6	a) Show that $p \leftrightarrow q$ and $(p \wedge q) \vee (\neg p \wedge \neg q)$ are logically equivalent b) Find post and pre order traversal from the given tree below <pre> 5 / \ 4 8 / \ / \ 11 18 / \ / \ 44 31 58 </pre> c) Suppose, you have 2 dataset of integer numbers. Write a C++ program that can be used to find the duplicate elements from these datasets and count the total numbers of such occurrences.	3 2 5
7	a) Let, Kuddus Ali and Co wants to purchase a software for his restaurant, named Kuddus Khanapina. So he decided to purchase it from a software company. The developers of the company called a meeting and discussed their cost and time to deliver the product. Discuss how the developers can use COCOMO model to compute the cost of software development and offer it to Kuddus Ali and Co. b) Consider a byte addressable system with a 32-bit virtual address space and a page size of 4 KB. If each page table entry (PTE) consumes 4 bytes, what is the total size of the page table? c) Draw a class diagram showing a hospital management system.	4 3 3
8	a) Suppose, you are working in a company's latest salary related data. There are around 10000 employees and the file you are having has the salary data in random order. The manager wants you to find some employees data and give it in a separate file. As a Computer Science graduate you decided to solve this problem using a C++ program. Now, discuss with code and proper steps which algorithm you will use while solving this issue. b) Write corresponding test name that matches with the examples given below	

		<table><tr><th>Details</th><th>Test Name</th></tr><tr><td>For an online banking system, 5,000 users simultaneously log in and check their account balances. Performance testing checks whether the system responds within an acceptable time.</td><td></td></tr><tr><td>After deploying a new version of an e-commerce website, testers check: <i>Website opens, User can log in, Products can be searched Product can be added to cart, Checkout page opens.</i> If these basic functions work, the build can proceed to detailed testing.</td><td></td></tr><tr><td>A banking application normally supports 10,000 simultaneous users. Testers gradually increase the load to 20,000, 30,000, or more users to determine when the system becomes slow or crashes.</td><td></td></tr><tr><td>For a mobile banking application, testers check whether: Transactions work correctly, The application is easy to use, User information is secure, The application does not frequently crash. Transactions are completed within an acceptable time</td><td></td></tr><tr><td>A developer modifies the payment module of an e-commerce website. Testers then check not only payment but also: Login, Cart, Product search, Order placement, Order history. This ensures that the new payment changes have not broken existing functionality.</td><td></td></tr></table>	Details	Test Name	For an online banking system, 5,000 users simultaneously log in and check their account balances. Performance testing checks whether the system responds within an acceptable time.		After deploying a new version of an e-commerce website, testers check: <i>Website opens, User can log in, Products can be searched Product can be added to cart, Checkout page opens.</i> If these basic functions work, the build can proceed to detailed testing.		A banking application normally supports 10,000 simultaneous users. Testers gradually increase the load to 20,000, 30,000, or more users to determine when the system becomes slow or crashes.		For a mobile banking application, testers check whether: Transactions work correctly, The application is easy to use, User information is secure, The application does not frequently crash. Transactions are completed within an acceptable time		A developer modifies the payment module of an e-commerce website. Testers then check not only payment but also: Login, Cart, Product search, Order placement, Order history. This ensures that the new payment changes have not broken existing functionality.		
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9	a) Discuss the difference between Dynamic Programming and Divide and Conquer techniques b) Discuss which CPU scheduling technique will be appropriate between Round robin (slice time 2ms) and Shortest job first while solving for following data <table><tr><th>Process</th><th>CPU Burst Times in ms</th></tr><tr><td>P1</td><td>8</td></tr><tr><td>P2</td><td>10</td></tr><tr><td>P3</td><td>6</td></tr><tr><td>P4</td><td>4</td></tr></table> c) Discuss different types of inheritance with proper codes	Process	CPU Burst Times in ms	P1	8	P2	10	P3	6	P4	4	3 3 4			
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10	a) Discuss ACID properties of database transaction b) Discuss how page fault can be handled in OS. Discuss some reasons for page fault. c) Discuss LRU and OPR techniques for the following data with page no. 3 1, 2, 3, 1, 2, 4, 1, 2, 5	3 4 3													